Tree-Based Machine Learning for Heart Disease Classification: A Comparative Study with Clinical Implications

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Overview of the data

This data set spans from May 1981 to September 1984 and comprises individual-level patient data from hospitals in Cleveland, Hungary, Switzerland, and Long Beach. The population of interest is existing patients in the hospital system. Therefore, models are expected to have high internal but lower external validity.

The data was collected by researchers who were interested in applying an algorithm to test different combinations of features that were available in different clinical settings (Detrano et al., 1989). Specifically, they ran 352 separate logistic regressions to account for different subsets of predictors. This project aims to improve upon the previous classification model through the application of tree-based machine learning algorithms. Following the suggestions of Detrano et al., this analysis uses the "best" subset of 13 features they identified in their research that are ubiquitous in clinical settings:

feature	description	data.type
age	age in years	numerical
sex	1 = male; 0 = female	categorical
cp	chest pain type (0 = typical angina; 1 = atypical angina; 2 = non-anginal pain; 3 = asymptomatic (no chest pain))	categorical
trestbps	resting blood pressure (in mm Hg on admission to the hospital)	numerical
chol	serum cholesterol in mg/dl	numerical
fbs	fasting blood sugar > 120 mg/dl (1 = true; 0 = false)	categorical
restecg	resting electrocardiographic results	categorical
thalach	maximum heart rate achieved	numerical
exang	exercise induced angina (1 = yes; 0 = no)	categorical
oldpeak	ST depression induced by exercise relative to rest	numerical
slope	the slope of the peak exercise ST segment	categorical
ca	number of major vessels colored by fluoroscopy	numerical
thal	thalassemia (0 = no record; 1 = normal; 2 = fixed defect; 3 = reversible defect)	categorical
target	the presence of heart disease in the patient (1 = yes; 0 = no)	categorical

These predictors are expected to be crucial for this classification problem. Among them, the number of colored vessels by fluoroscopy (ca) and chest pain type are expected to have the strongest signal. The first feature indicates whether a heart blockage exists, in which less colored vessels in imaging tests could suggest poor blood flow. The latter is a strong signal of heart disease in a medical context. Specifically, typical angina, a chest pain induced by reduced blood flow to the heart that follows the classic pattern of symptoms, is likely to be a strong signal for heart disease.

By the end of the project, the final predictive model is expected to act as a clinical support tool to inform personalized medical decisions regarding heart disease. Doctors and hospitals could use the model to predict

potential heart disease patients and provide necessary medical supports, which not only enhances resource allocation but is also economically efficient as medical procedures like invasive imaging tests are expensive.

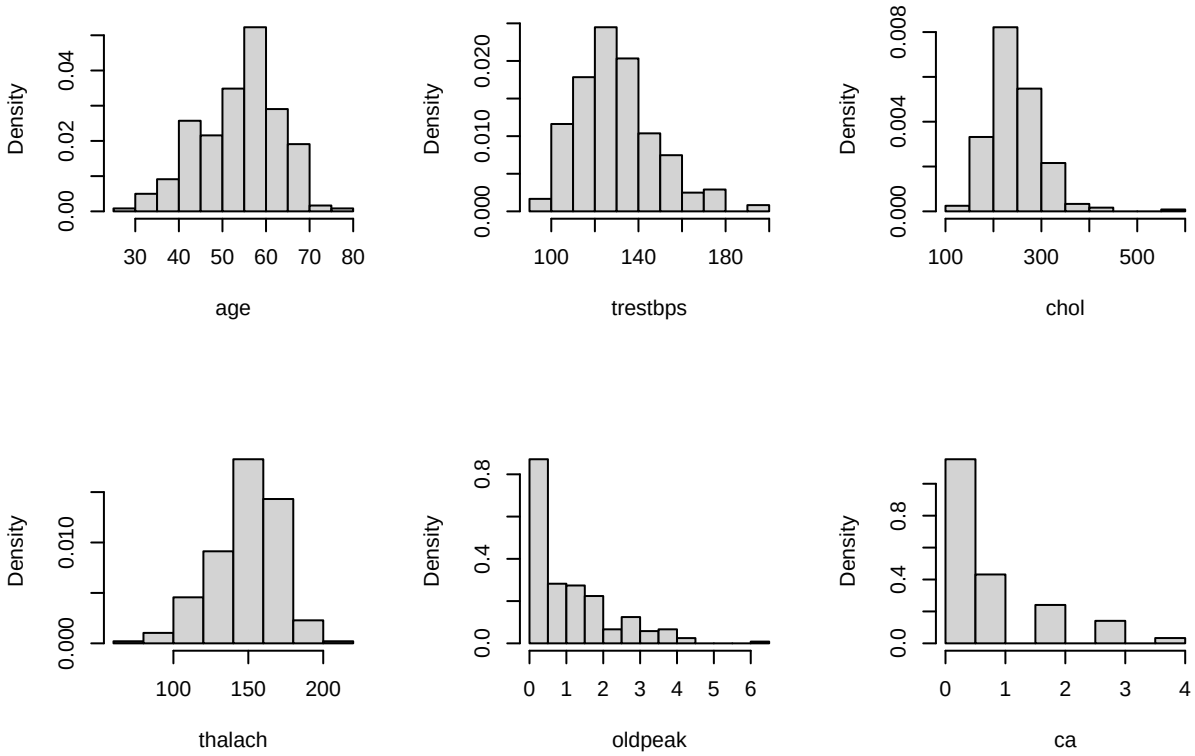
We have adopted 80/20 train-test split, and the exploratory data analysis (EDA) and modeling will be performed on the training set.

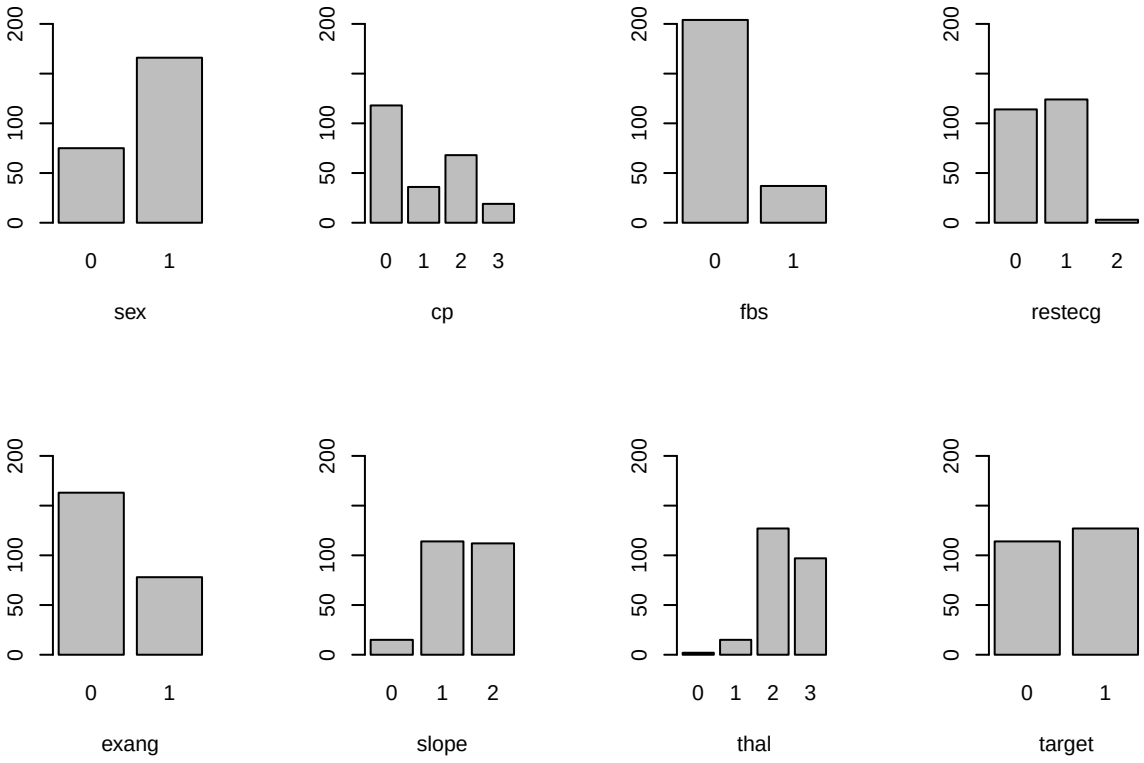
Exploratory Data Analysis

Summary Statistics

	mean	median	std.dev	maximum	minimum	range
age	54.3609959	56.0	9.135461	77.0	29	48.0
trestbps	132.4315353	130.0	18.494945	200.0	94	106.0
chol	245.4605809	240.0	52.641234	564.0	126	438.0
thalach	149.2531120	152.0	22.217819	202.0	71	131.0
oldpeak	1.0780083	0.8	1.181866	6.2	0	6.2
ca	0.7344398	0.0	1.030785	4.0	0	4.0

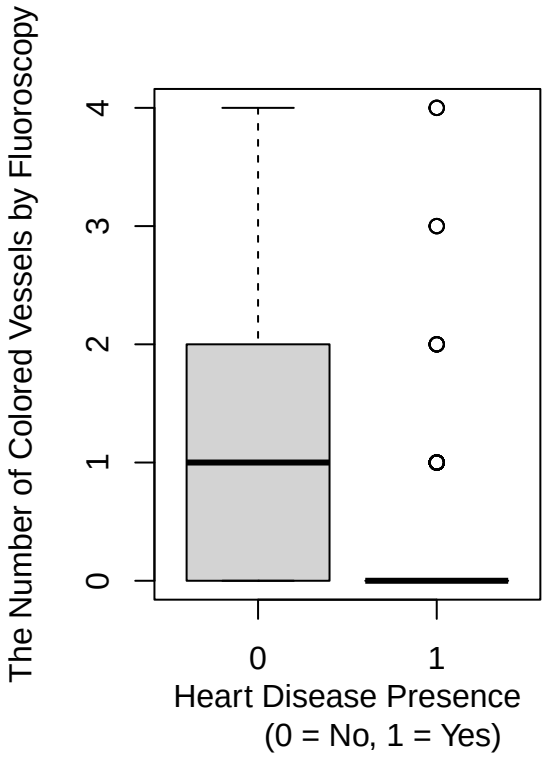
	HeartDiseaseYes	HeartDiseaseNo	NA.	Mode
the presence of heart disease	127	114	0	1

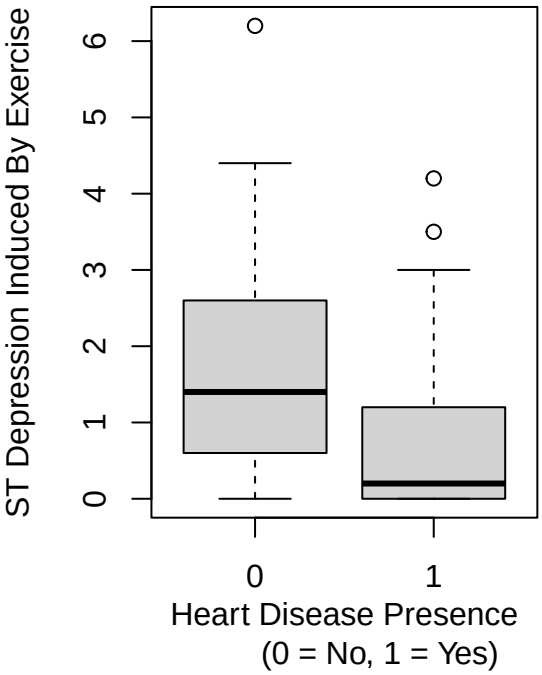




The bar-plots reveal the proportion of each categorical feature. It is worth noting that our target, the presence of heart disease, is balanced, which suggests that we can use ROC curve and its area-under-curve to measure model performance.

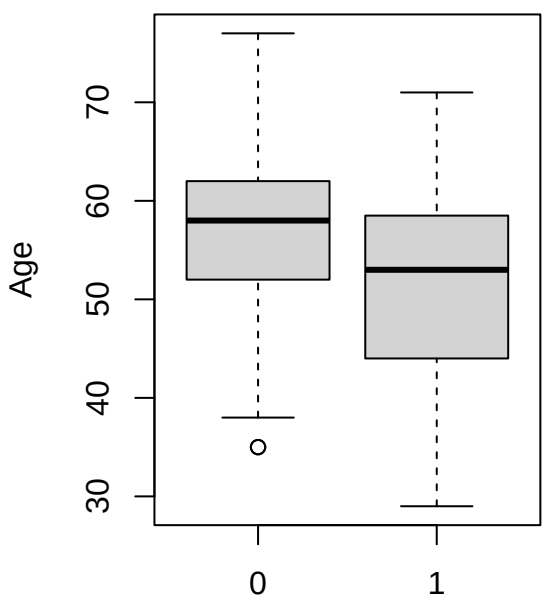
Bivariate Analysis

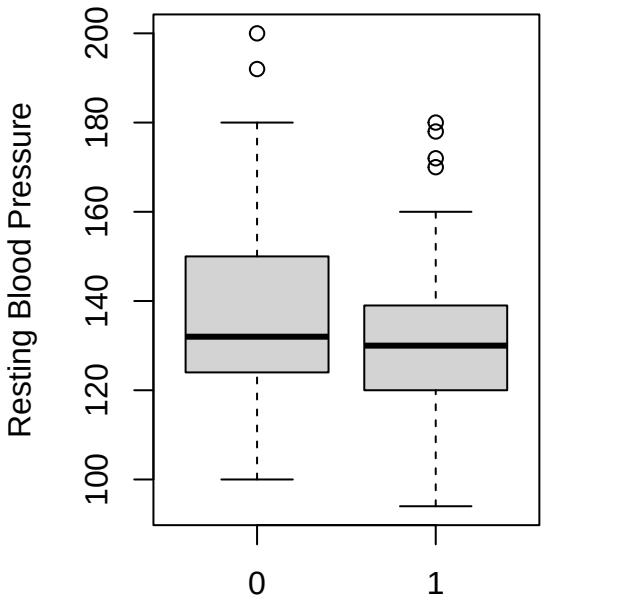





```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  train.heart$ca by train.heart$target
## W = 10618, p-value = 2.417e-12
## alternative hypothesis: true location shift is not equal to 0
```

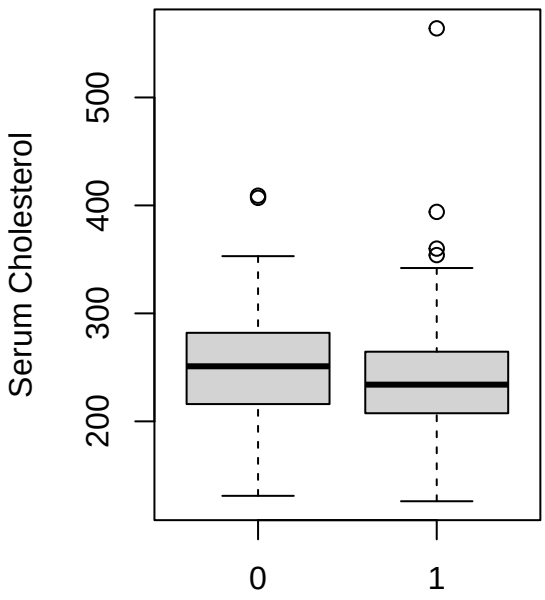
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  train.heart$oldpeak by train.heart$target
## W = 10618, p-value = 2.074e-10
## alternative hypothesis: true location shift is not equal to 0
```

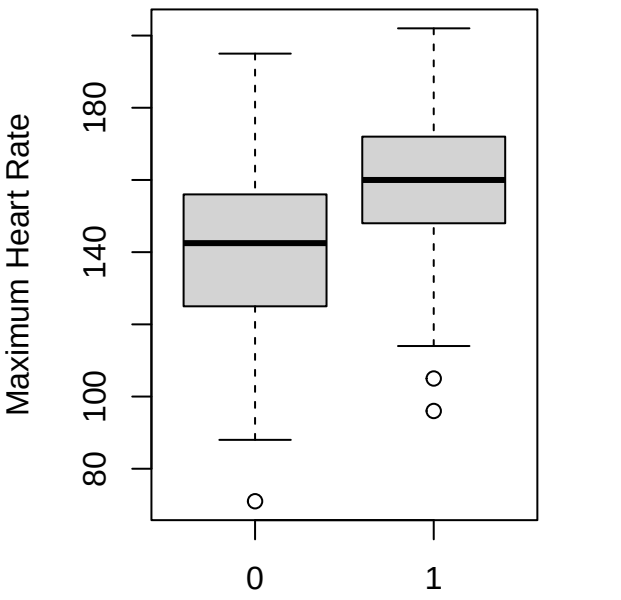





```
##
##  Welch Two Sample t-test
##
## data:  train.heart$age by train.heart$target
## t = 3.9218, df = 238.79, p-value = 0.0001149
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  2.218978 6.698138
## sample estimates:
## mean in group 0 mean in group 1
##        56.71053        52.25197

##
##  Wilcoxon rank sum test with continuity correction
##
## data:  train.heart$trestbps by train.heart$target
## W = 8615.5, p-value = 0.0107
## alternative hypothesis: true location shift is not equal to 0
```

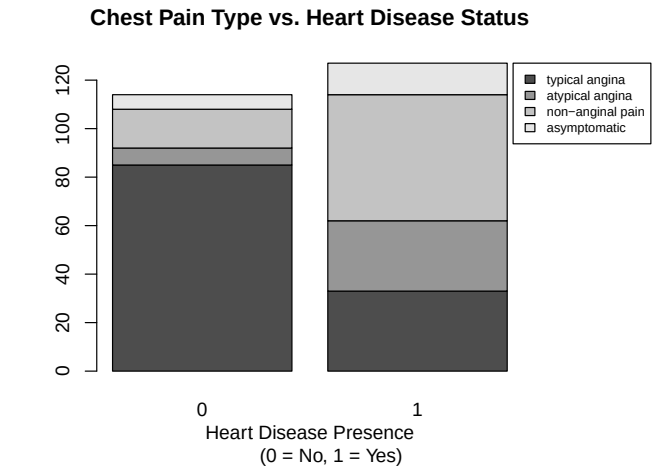


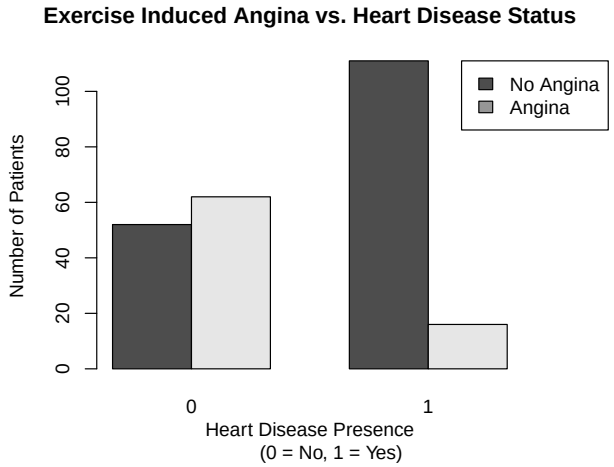



```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  train.heart$chol by train.heart$target
## W = 8415.5, p-value = 0.02952
## alternative hypothesis: true location shift is not equal to 0
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  train.heart$thalach by train.heart$target
## W = 3822, p-value = 2.549e-10
## alternative hypothesis: true location shift is not equal to 0
```

From the results of Wilcoxon Rank Sum Tests and Two-Sample t-test for age, it can be identified that all numerical features are statistically different ($p < 0.05$) at the 5% significance level between patients with heart disease and those without, which signify their high potential predictive power.






```
##
##  Pearson's Chi-squared test
##
## data:  cp.table
## X-squared = 57.463, df = 3, p-value = 2.046e-12

##                Presence of Heart Disease
## Chest Pain Type  0  1
##               0 85 33
##               1  7 29
##               2 16 52
##               3  6 13

##
##  Fisher's Exact Test for Count Data
##
## data:  thal.table
## p-value < 2.2e-16
## alternative hypothesis: two.sided
```

```
##            Presence of Heart Disease
## Thalassemia   0   1
##           0   1   1
##           1  11   4
##           2  27 100
##           3  75  22
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  sex.table
## X-squared = 15.171, df = 1, p-value = 9.82e-05
```

```
##    Presence of Heart Disease
## Sex  0  1
##   0 21 54
##   1 93 73
```

```
##
##  Pearson's Chi-squared test
##
## data:  slope.table
## X-squared = 38.714, df = 2, p-value = 3.921e-09
```

```
##      Presence of Heart Disease
## Slope  0  1
##     0  9  6
##     1 76 38
##     2 29 83
```

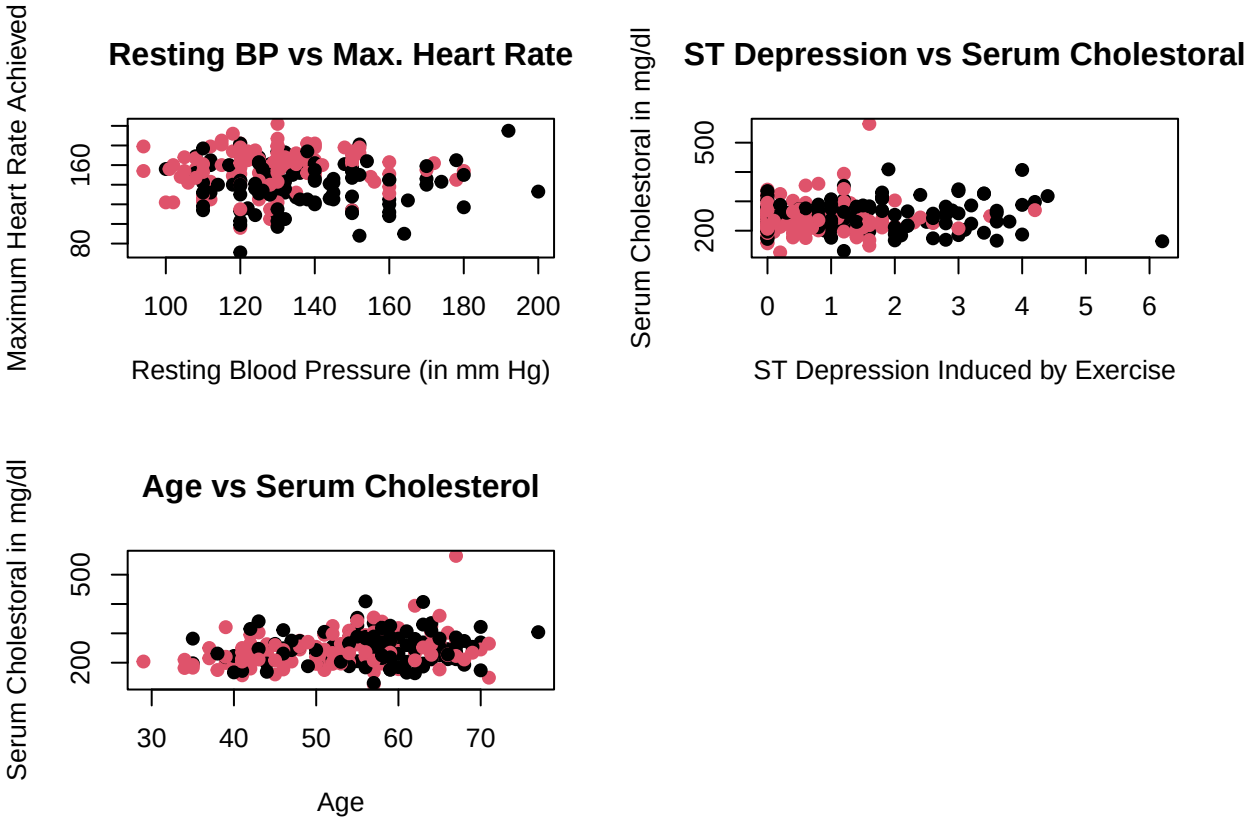
```
##
##  Fisher's Exact Test for Count Data
##
## data:  restecg.table
## p-value = 0.002117
## alternative hypothesis: two.sided
```

```
##                                    Presence of Heart Disease
## Resting Electrocardiographic Results  0  1
##                                    0 64 50
##                                    1 47 77
##                                    2  3  0
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  angina.table
## X-squared = 46.032, df = 1, p-value = 1.163e-11
```

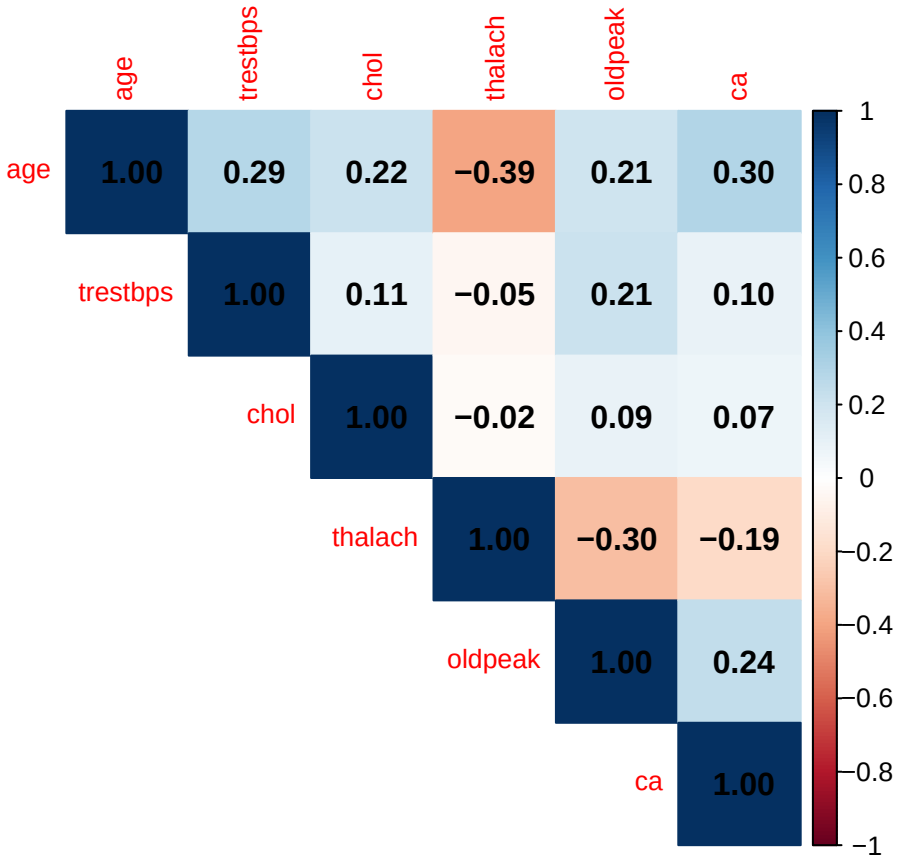
```
##                         Presence of Heart Disease
## Exercise Induced Angina?   0   1
##                        0  52 111
##                        1  62  16
```

The results of χ^2 Tests and Fisher's Exact Tests ($p < 0.05$) suggest statistically significant association between these features and the presence of heart disease at the 5% significance level. Thus, these features could have high predictive power.



These scatter-plots enable us to assess class separability. We can see the data points seem well-separate, which suggests that these feature pairs could be strong predictors for the model.

Correlation Heatmap

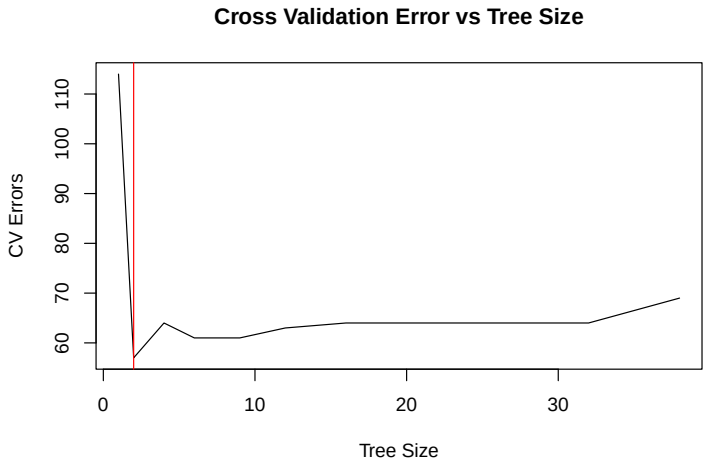


The correlation heatmap reveals the correlation and direction of relationships between numerical features using Pearson's r coefficient since we are examining continuous numerical features. In the bivariate analysis section, we have examined the predictive power of each feature. The primary focus of the heatmap is to identify redundancy among numerical predictors. The matrix suggests that the correlations are generally low, and thus, no obvious linear relationships exist. Therefore, each feature is expected to provide distinct clinical information to the classification process. In other words, none of these features is redundant in this subset of features. The weak linear correlations observed in the heatmap motivated the use of tree-based methods, which can capture nonlinear relationships and feature interactions beyond linear assumptions.

Conclusion

From the EDA, we have explored the data structure and possible relationships through statistical testing and visualization. The balanced nature of presence of heart disease is likely due to this specific population of interest: these patients are already suspected of having heart disease. Moreover, the results provide a broad picture of how each feature would influence prediction in the tree-based analysis. In summary, this feature subset from Detrano and his colleagues (1989) shows strong predictive potential.

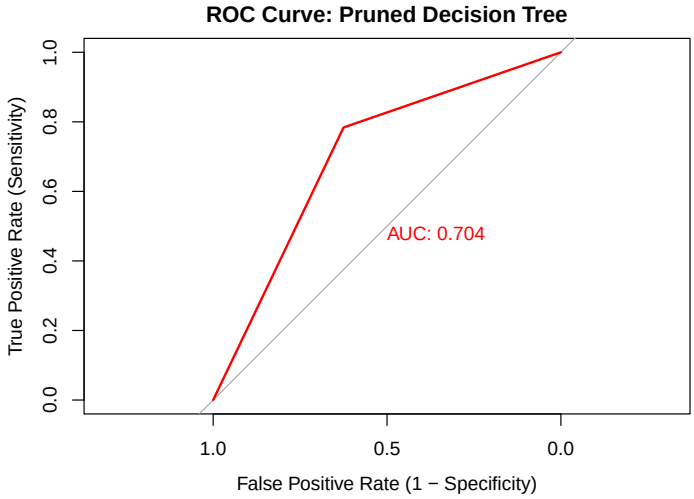
Plain Tree



In this analysis, the plain tree will act as the baseline of this classification problem. Firstly, we fit a tree and grow it as large as possible. Then, we use 10 fold cross-validation to select the optimal tree size. Lastly, we prune the tree with the chosen size.

Model Performance

ROC



threshold	specificity	sensitivity
0.5121218	0.625	0.7837838

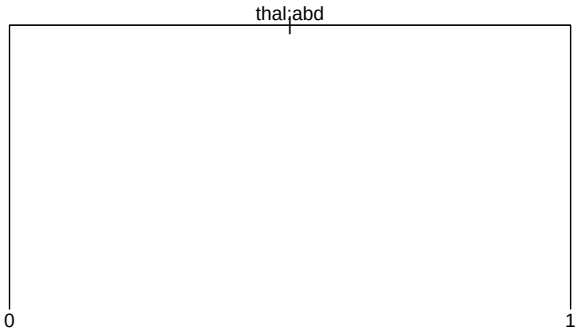
The ROC area under curve is 0.704, which suggests a fair model performance. The model performs better than random guessing, which would be roughly 0.5 based on the proportion of the heart disease presence in this sample, suggesting a moderate capability in distinguishing between heart disease and no heart disease.

At the best threshold around 0.51, true positive rate and false positive rate are 0.784 and 0.375, respectively. In this context, the model is expected to capture 78.4% of patients who actually have heart disease, and about 37.5% of patients without heart disease would be misclassified.

*This result will be the baseline performance to compare with.

Confusion Matrix and Test Error

Plain Tree Interpretation




```
## node), split, n, deviance, yval, (yprob)
##       * denotes terminal node
##
## 1) root 241 333.4 1 ( 0.4730 0.5270 )
##   2) thal: 0,1,3 114 124.8 0 ( 0.7632 0.2368 ) *
##   3) thal: 2 127 131.4 1 ( 0.2126 0.7874 ) *
```

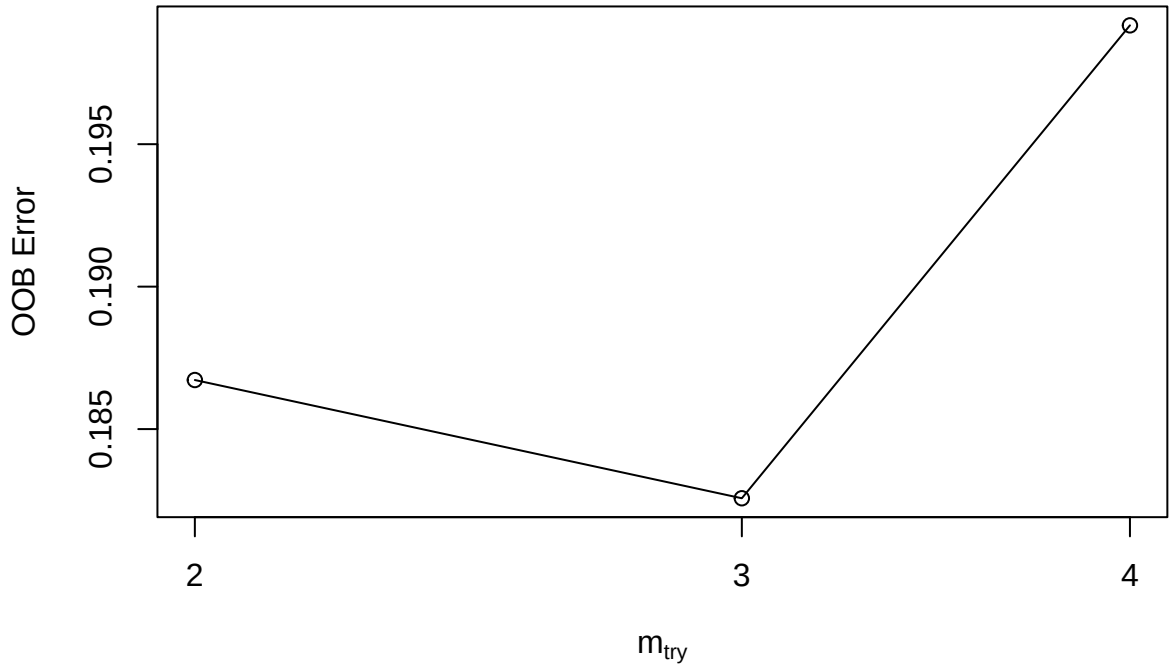
Using cross-validation, the "best" tree size is 2. As a result, our plain tree only has one split, which is on thalassemia. When thal = 2, the predicted class is 1 (disease). In the context, it means if a fixed defect presents, that patient would be predicted to have heart disease.

Random Forest

The pruned tree has high variance and low bias. Consequently, the model performance significantly decreases on unseen data. To address this issue, we leverage the random forest algorithm and manage bias-variance trade-off through aggregation. Specifically, we aggregate the result of many fully grown trees through majority vote to reduce the variance. We randomly select m features for each split to ensure trees are independent of each other in order to reduce model variance.

Since we introduce noise at each split and aggregate the result, the number of trees will not lead to overfitting; simultaneously, tree depth is less sensitive, for which we set minimum leaf node size to 10 to prevent extreme cases. The only tuning parameter is the number of features selected at each split in this model, and it will be chosen through examining the out-of-bag error.

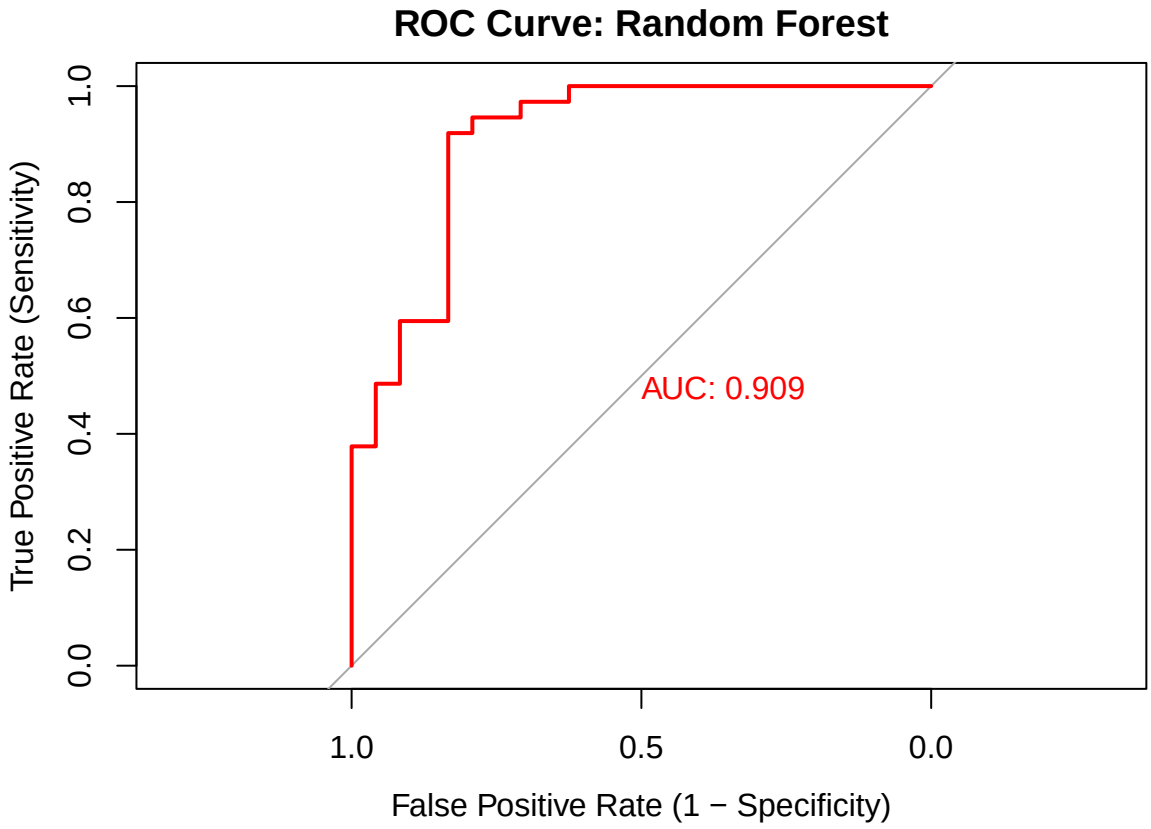
```
## mtry = 3  OOB error = 18.26%
## Searching left ...
## mtry = 2     OOB error = 18.67%
## -0.02272727 0.01
## Searching right ...
## mtry = 4     OOB error = 19.92%
## -0.09090909 0.01
```



The "best" number of features selected at each split is 3.

Model Performance

ROC



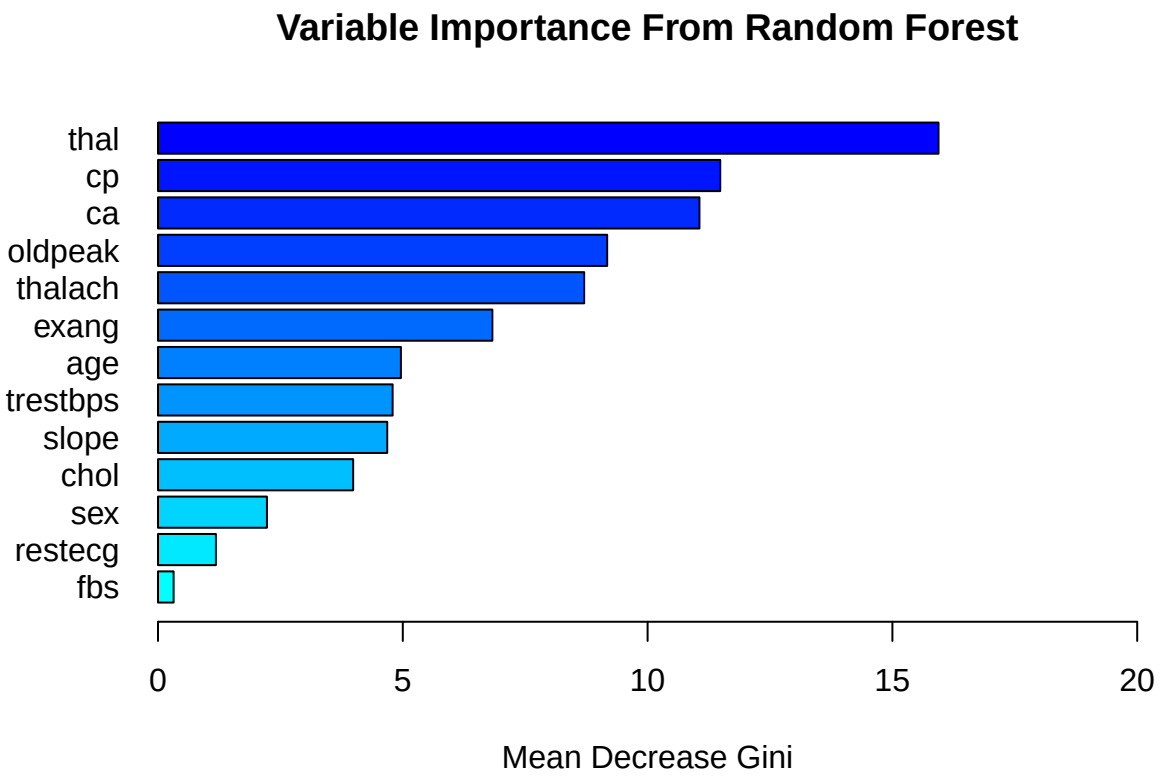
threshold	specificity	sensitivity
0.5065	0.8333333	0.9189189

The ROC area under curve is 0.909, which suggests a high model performance. Compared to the baseline regression tree, the performance of random forest is much better. At the best threshold, 0.5065, the true positive rate (sensitivity) and false positive rate (1 - specificity) are 0.92 and 0.17, respectively. Compared to the baseline, random forest is expected to capture a lot more patients who actually have heart disease, and it is also expected to have less misclassification.

Confusion Matrix and Test Error

Random Forest Interpretation

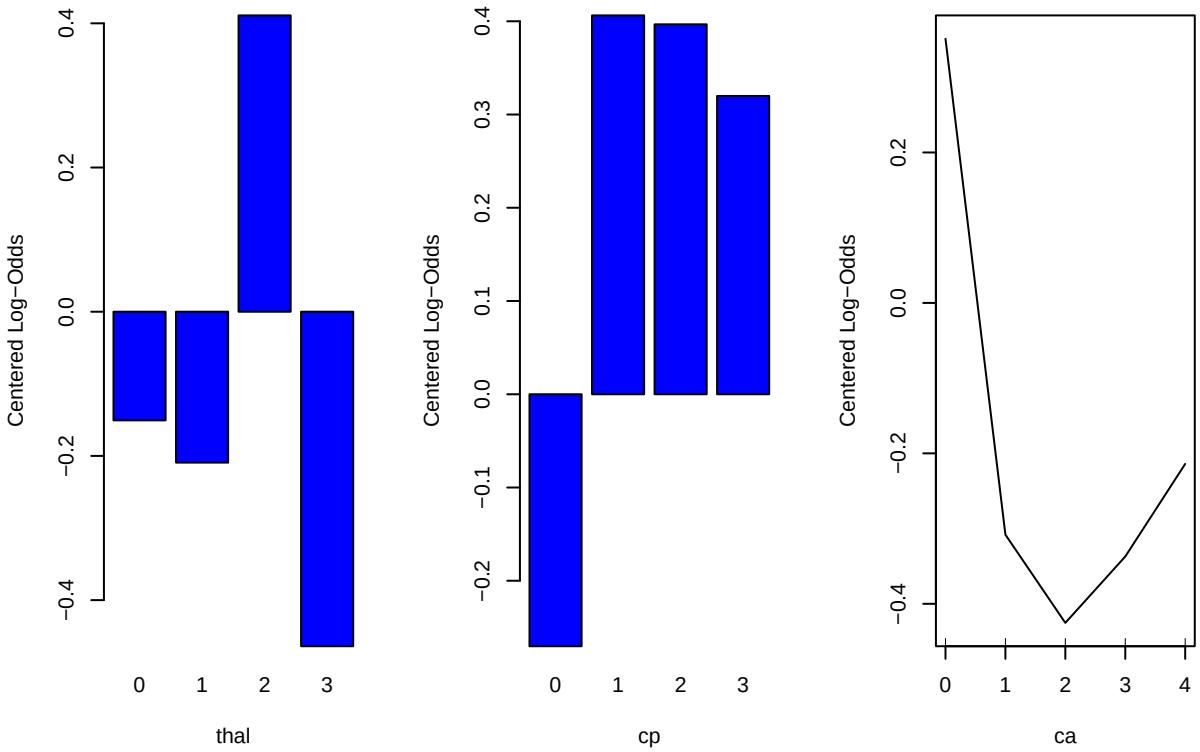
Variable Importance



feature	MeanDecreaseGini
thal	15.9415931
cp	11.4836872
ca	11.0585600
oldpeak	9.1742116
thalach	8.7041125
exang	6.8292946
age	4.9611742
trestbps	4.7921662
slope	4.6817248
chol	3.9857824
sex	2.2249866
restecg	1.1824824
fbs	0.3186016

From the horizontal bar-plot and table, we can tell thal (thalassemia) is the most important feature, which contributed the most to mean Gini index reduction, followed by cp (chest pain type) and ca (number of major vessels colored).

Partial Dependence Plot for Top 3



The partial dependence plots showcase the marginal effects of the top 3 features on whether a patient has heart disease (target = 1). The y-axis represents the log-odds. Roughly speaking, when the value of y is greater than 0, the feature increases the probability of having heart disease, and vice versa.

Coherent with the hypothesized relationships and the results of the EDA, the number of colored vessels by fluoroscopy and chest pain type are strong predictors. Additionally, thalassemia is found to be the most important feature.

The practical implications are:

(1) when the number of major vessels colored by fluoroscopy increases, where 0 implies blockage, we can see more healthy vessels. Intuitively, the more colored vessels we see, the less likely there is a blockage. Even though there is an uptick at ca = 2, the log-odds are still negative, meaning lower relative risk.

(2) when a patient exhibits fixed defect (thal = 2), defect does not change with stress or rest, it could indicate damaged heart muscle, and, therefore, patient would have a higher relative heart disease risk.

(3) The partial dependence plot suggests that typical angina has a negative log-odds, suggests lower relative risk of heart disease. The rest have positive log-odds, thereby high relative risk. It is worth noting that patients who do not experience chest pain during exercise testing (cp = 3) also have high relative risk. While it sounds counter-intuitive, the population of interest is indeed suspects of heart disease, and no symptoms could lead to delayed treatment. In contrast, typical chest pain (cp = 0) might not be a powerful predictive signal since everyone in this population of interest is already a suspected heart disease patient.

Boosted Tree

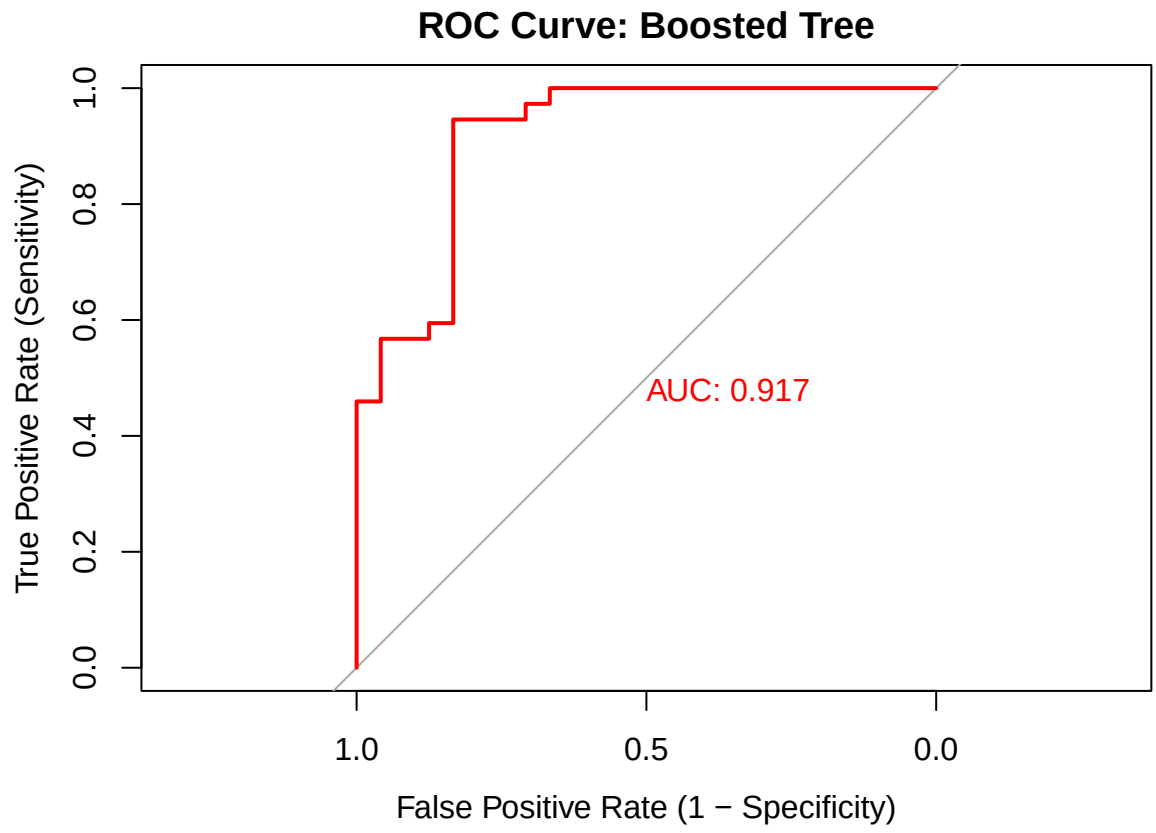
While random forest provides a strong performance, the performance of another powerful approach of ensemble method, boosting, is also of interest in this analysis. This first-order gradient boosting method could outperform modern second-order approach in relatively smaller sample size such as this data set. It is because this classic boosting approach is less prone to overfit.

There are multiple tuning parameters, learning rate, interaction depth, minimum terminal node size, and the number of trees, and these are tuned through grid search approach.

	interaction.depth	learning.rate	min.obs.in.node	optimal.tree	cv.deviance
10	2	0.001	20	4964	0.7833541

Model Performance

ROC



threshold	specificity	sensitivity
0.4770031	0.8333333	0.9459459

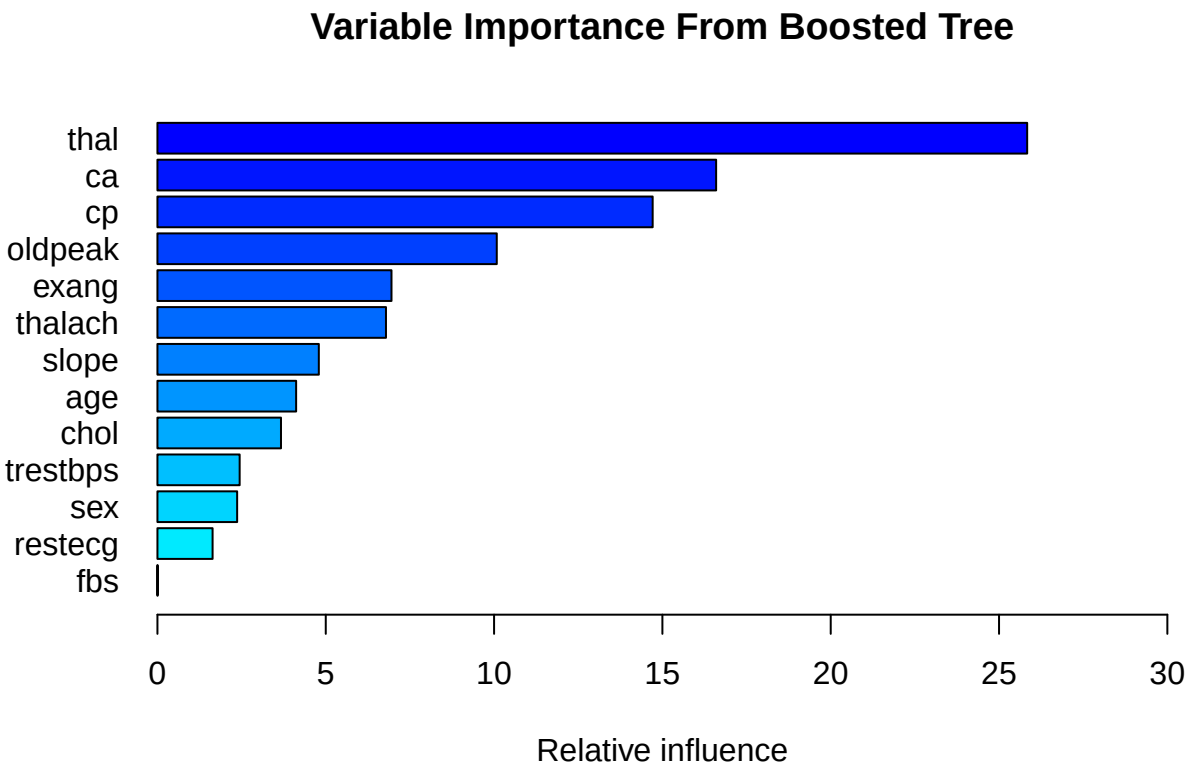
The ROC area under curve is 0.917, which suggests that a high model performance. The boosted tree significantly outperforms pruned tree in terms of AUC, and it also performs slightly better than random forest with higher sensitivity. At the best threshold, 0.426, the true positive rate (sensitivity) and false

positive rate (1 - specificity) are 0.95 and 0.17, respectively. This result suggests that the boosted tree is expected to capture a lot more patients who actually have heart disease and have less misclassification compared to pruned tree and random forest.

Confusion Matrix and Test Error

Boosted Tree Interpretation

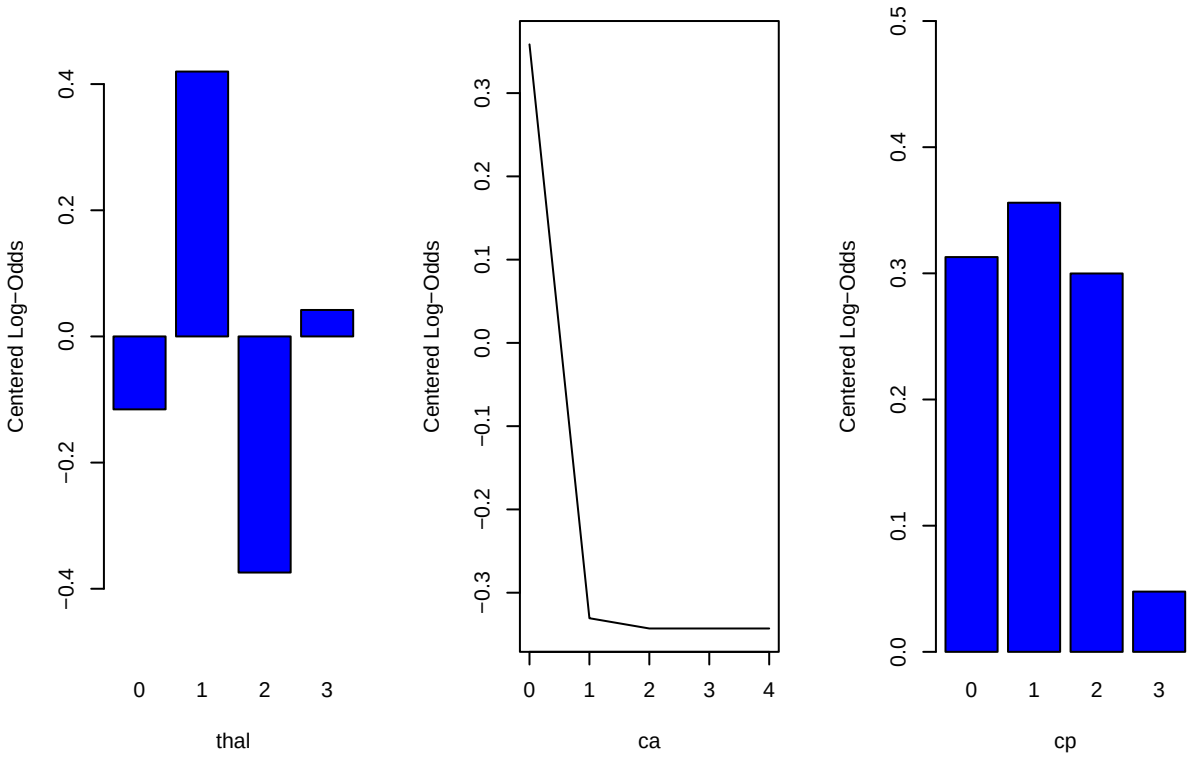
Variable Importance



	var	rel.inf
thal	thal	25.8358275
ca	ca	16.5959354
cp	cp	14.7106326
oldpeak	oldpeak	10.0798214
exang	exang	6.9523656
thalach	thalach	6.7885039
slope	slope	4.7941376
age	age	4.1199323
chol	chol	3.6699491
trestbps	trestbps	2.4421823
sex	sex	2.3676342
restecg	restecg	1.6383426
fbs	fbs	0.0047357

Examination of feature importance in the boosted tree shows that thalassemia contributed the most to predict the presence of heart disease, following by the number of major vessels (0-3) colored by fluoroscopy and chest pain type.

Partial Dependence Plot for Top 3 features



The partial dependence plots showcase the marginal effects the top 3 variables have on the boosted tree's prediction.

The results of the boosted tree align with the hypothesized relationships and the results of the EDA. Like random forest, Thalassemia, the number of colored vessels by fluoroscopy, and chest pain type appear to be the most important features in the boosted tree. Notably, the partial dependence plots differ from those of the random forest, which implies that these two approaches see the data differently and learn different patterns due to the fundamentally different mechanisms.

The practical implications are:

(1) Patients with fixed defect (thal = 2) and patients who do not have thalassemia record (thal = 0) correspond to negative centered log-odds, which suggests that the probability of heart disease presence among these patients are lower compared to the cohort mean, whereas patients who are normal (thal = 1) have the highest relative risk.

(2) when there is no colored vessel by fluoroscope, it could indicate a blockage, the partial dependence suggests the highest relative risk.

(3) Patients with any type of angina (chest pain) correspond to positive log-odds. Even when it is asymptomatic (cp = 3), it still corresponds with positive log-odds, which is likely due to the unique population of interest: they are all heart disease suspected patients.

XGBoost

As mentioned in the boosting section, second order gradient boosting might not yield better performance in a relatively small sample size. It is because the Hessian information matrix, the second derivative, could be noisy, thereby prone to overfit. We would like to test if this is the case if we incorporate proper LASSO and Ridge regularization. In this section, we will fit a very powerful tree based model, XGBoost, which has dominated Kaggle competitions for years. For tuning, we will use AUC as the evaluation metric as we are more interested in the separability of the model than the predicted probability. We will also use Bayesian Optimization to find the global optimal parameters. We will tune 8 different hyperparameters:

1) learning rate
2) maximum depth of a tree
3) minimum sum of instance weight needed in a child
4) the subsample ratio of training data
5) the subsample ratio of columns when constructing each tree
6) L1 (LASSO) regularization
7) L2 (Ridge) regularization
8) Minimum loss reduction required to make a further split

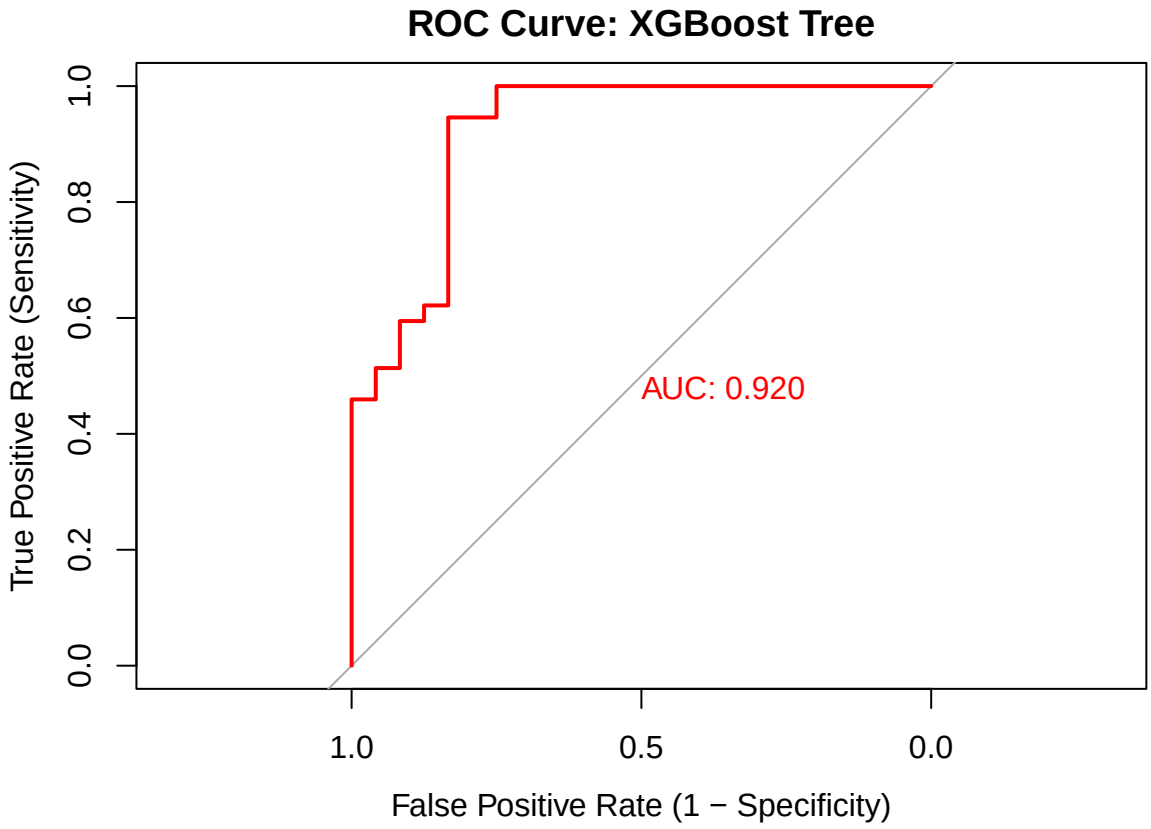
In the meantime, the Bayesian Optimization algorithm will identify the optimal number of boosting iteration. Since we have 8 tuning parameters, the number of random parameter combinations we start with is 40 based on the convention, multiplying 5 by the number of tuning parameters, and the number of combinations chosen by the Gaussian Process by learning from previous results is 80, which is usually 2 times the random parameter combinations we start with.

eta	max_depth	min_child_weight	subsample	colsample_bytree	gamma	lambda	alpha
0.1667002	7	2.633753	0.6876178	0.8048109	1.331869	1.490141	2.806211

- The optimal iteration identified by Bayesian Optimization is 64.

Model Performance on Test Set

ROC



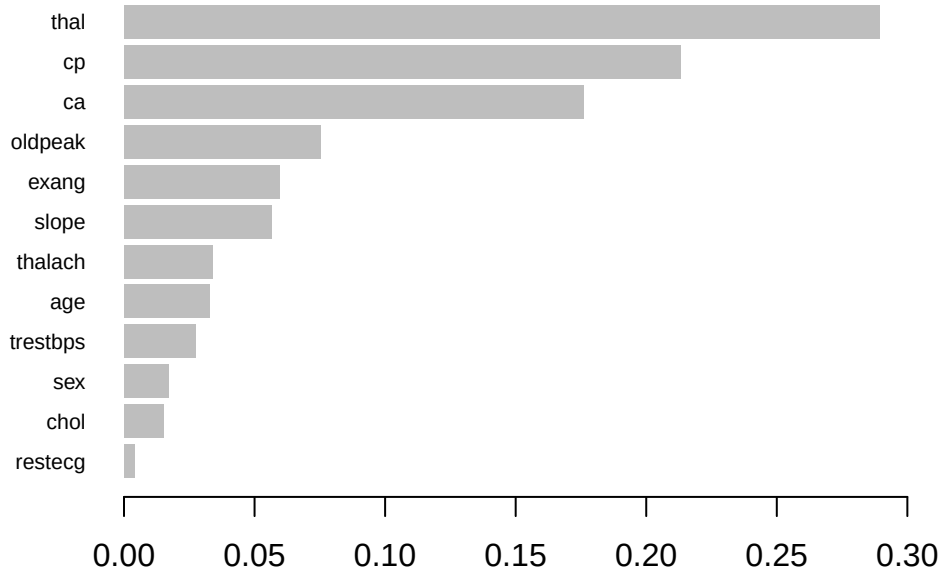
threshold	specificity	sensitivity
0.4633494	0.8333333	0.9459459

The ROC-AUC is 0.92, which is similar to boosting but slightly better than random forest. Compared to the AUC of the baseline, it is a significant improvement. At the best threshold, 0.46, the true positive rate (sensitivity) and false positive rate (1 - specificity) are 0.95 and 0.17, respectively. This result is identical to first order gradient boosting, which suggests that the model is expected to capture 95% of patients who actually have heart disease, and about 17% of patients without heart disease would be misclassified.

Confusion Matrix and Test Error

XGBoost Interpretation

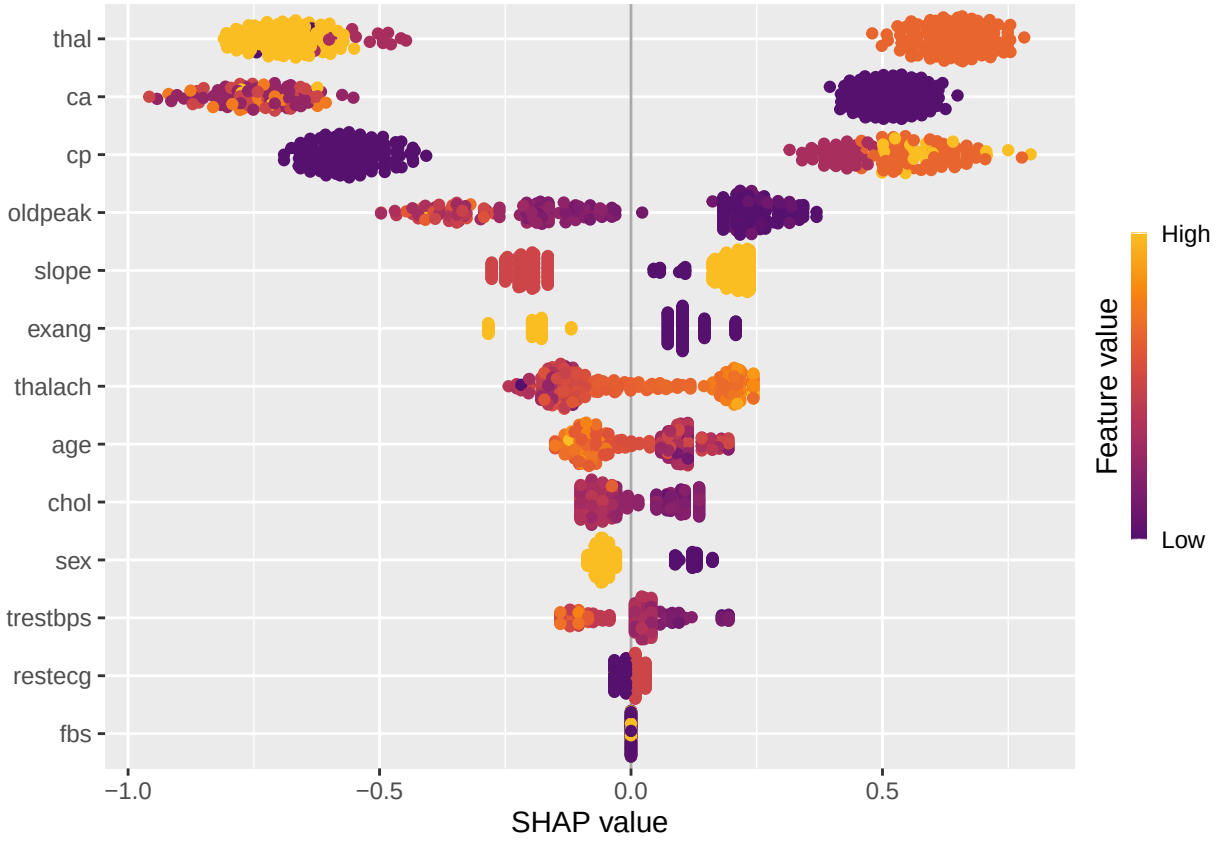
Variable Importance



Feature	Gain	Cover	Frequency
thal	0.2893195	0.1870928	0.1388889
cp	0.2131655	0.1673525	0.1527778
ca	0.1759776	0.1853369	0.1805556
oldpeak	0.0751541	0.1139754	0.1250000
exang	0.0596520	0.0528409	0.0555556
slope	0.0566123	0.0721277	0.0694444
thalach	0.0340342	0.0718028	0.0694444
age	0.0325574	0.0364917	0.0555556
trestbps	0.0272485	0.0455160	0.0555556
sex	0.0169528	0.0303384	0.0416667
chol	0.0152115	0.0303234	0.0416667
restecg	0.0041148	0.0068014	0.0138889

Similar to other models, thalassemia is the most important feature, following by the number of colored vessels by fluoroscope and chest pain type.

Shapley Values



In the SHAP plot, each dot represents one observation, and the association is specific to how the model sees the data. For thalassemia, reversible defect (thal = 3, yellow), no record (thal = 0, dark purple), and normal (thal = 1) are associated with negative Shapley values, whereas fixed defect (thal = 2) is associated with positive Shapley values. This result suggests that having fixed defect increases the probability of heart disease.

Conclusion

In summary, we have developed a pruned tree, random forest, first order gradient boosted tree, and XGBoost for this heart disease data set. Thalassemia (thal), the number of major vessels colored by fluoroscopy (ca), and chest pain type (cp) are consistently the top 3 features across all advanced models, which suggests these three predictors have a strong heart disease predictive signal.

The data were collected by Dr. Robert Detrano and his colleagues (1989), and this baseline paper had set a solid foundation for research and tool-building as it established the 14-feature subset as a standard diagnostic framework. This project built upon this benchmark using tree based methods. The expected outcome aims to act as a clinical supporting tool which informs personalized medical decisions among existing patients in the hospital system regarding heart disease.

We adopted an 80/20 train-test split and evaluated the model performance using ROC curve and its area-under-curve since classes (0/1) are roughly balanced. To prevent overfitting, all tuning parameters are chosen through 5 or 10 fold CV, out-of-bag error CV, Grid Search CV, or Bayesian Optimization, ensuring robustness and generalization.

Model Comparison

Performance on Test Data

Table 13: Pruned Tree

Test.Error.	FPR.	FNR.	TPR.	TNR.
0.2786885	0.375	0.2162162	0.7837838	0.625

```
##       predicted
## actual  0  1
##      0 15  9
##      1  8 29
```

Table 14: Random Forest

Test.Error.	FPR.	FNR.	TPR.	TNR.
0.1147541	0.1666667	0.0810811	0.9189189	0.8333333

```
##       predicted
## actual  0  1
##      0 20  4
##      1  3 34
```

Table 15: Boosted Tree

Test.Error.	FPR.	FNR.	TPR.	TNR.
0.0983607	0.1666667	0.0540541	0.9459459	0.8333333

```
##       predicted
## actual  0  1
##      0 20  4
##      1  2 35
```

Table 16: XGBoost

Test.Error.	FPR.	FNR.	TPR.	TNR.
0.0983607	0.1666667	0.0540541	0.9459459	0.8333333

```
##       predicted
## actual  0  1
##      0 20  4
##      1  2 35
```

Theoretical Performance at the "Best" Threshold

Table 17: Pruned Tree

threshold	specificity	sensitivity
0.5121218	0.625	0.7837838

Table 18: Random Forest

threshold	specificity	sensitivity
0.5065	0.8333333	0.9189189

Table 19: Boosted Tree

threshold	specificity	sensitivity
0.4770031	0.8333333	0.9459459

Table 20: XGBoost

threshold	specificity	sensitivity
0.4633494	0.8333333	0.9459459

Comparing the model performance, it can be identified that the advanced tree-based methods yield a significant improvement upon the plain tree. As expected, the pruned tree has high variance, and its performance on unseen data is undesirable, whereas other advanced tree-based methods manage to achieve a better bias-variance trade-off and yield better results. It is worth noting that the performance of boosted tree (first order gradient boosting) and XGBoost (second order gradient boosting) is almost identical on test data, which aligns with the expectation that Hessian Information Matrix is somewhat noisy in a relatively small sample size, thereby learning from this additional piece of information provides limited benefit.

Recommendation

While all advanced models showcase the exceptional predictive performance, the boosted tree and XGBoost have the lowest false negative rate. This is more important than false positive rate in clinical settings because a false negative means a patient who has heart disease is misclassified as no heart disease, which leads to no treatment. The severe repercussions from false negatives include but are not limited to heart attack and stroke. Among these two models, boosted tree is preferred given the relatively small sample size. The XGBoost model converged at a high learning rate with low iteration, which suggests a potential risk of stability, whereas the boosted tree learns from the data gradually over many iterations.

We can also make a trade-off between false positive rate and false negative rate by adjusting the threshold to better align with the nature of expensive false negative in clinical settings. By lowering the threshold, the model would be more sensitive to capture more true heart disease presence at the expense of more false positive.

```
##       predicted
## actual  0  1
##      0 16  8
##      1  0 37
```

Test.Error.	FPR.	FNR.	TPR.	TNR.
0.1311475	0.3333333	0	1	0.6666667

At threshold ≈ 0.2567, sensitivity = 1 and specificity = 0.625 based on the ROC curve, which suggests that at this threshold, we are expected to have no false negatives but 37.5% of patients without heart disease are expected to be misclassified theoretically. On the test set, the false positive and false negative rate are 0.33 and 0, which aligns with the purpose of adjusting the threshold.

References

Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J. J., Sandhu, S., Guppy, K. H., Lee, S., & Froelicher, V. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. The American journal of cardiology, 64(5), 304–310. https://doi.org/10.1016/0002-9149(89)90524-9